

popquiz acceleration, circular motion

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:55pm

Quiz Instructions

bonus points ahoy



Question 1 25 pts

In a uniform circular motion

hint: force, acceleration, displacement, velocity are vectors



The acceleration is constant



The centripetal force is constant



The velocity is constant



The speed is constant



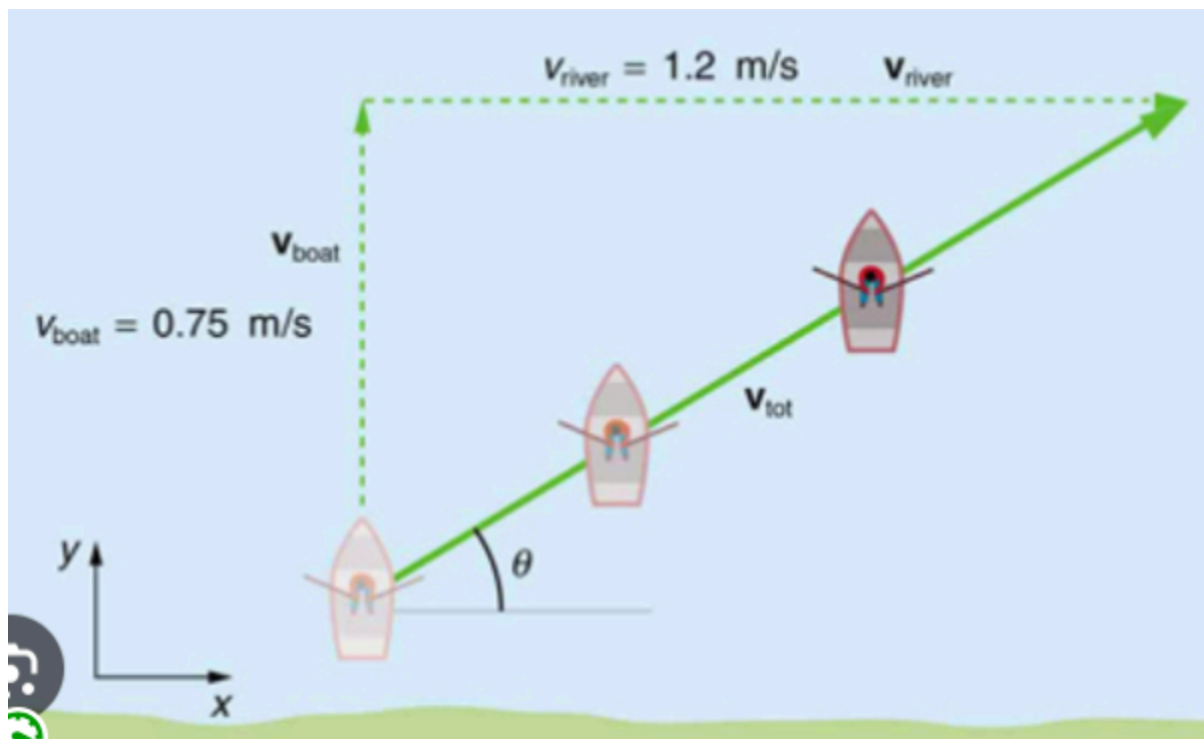
Question 2 20 pts

In a circular uniform motion, the acceleration

- ☐ point toward the center of the circle
- ☐ It curves around the circle
- ☐ there is no acceleration in such motion.
- ☐ is tangent to the circle



Question 3 20 pts



The green vector shows the velocity (a vector) of the boat relative to the shore. It is the the vector sum of the velocity of the boat relativet to the river (V_x) and the velocity of the river relative to shore (V_y)

What is the standard notion for the velocity of the boat relative to shore? (in bold green). That is find magnitude and direction relative to the positive xax-is

hint: Attach the green vector to the origin of a (X,Y) system. Connect the origin (tail) to (V_x, V_y) or head.

- ☐ 1.4m/s @58 / x-axis
- ☐ 2 m/s @ 32 /+x-axis

☐

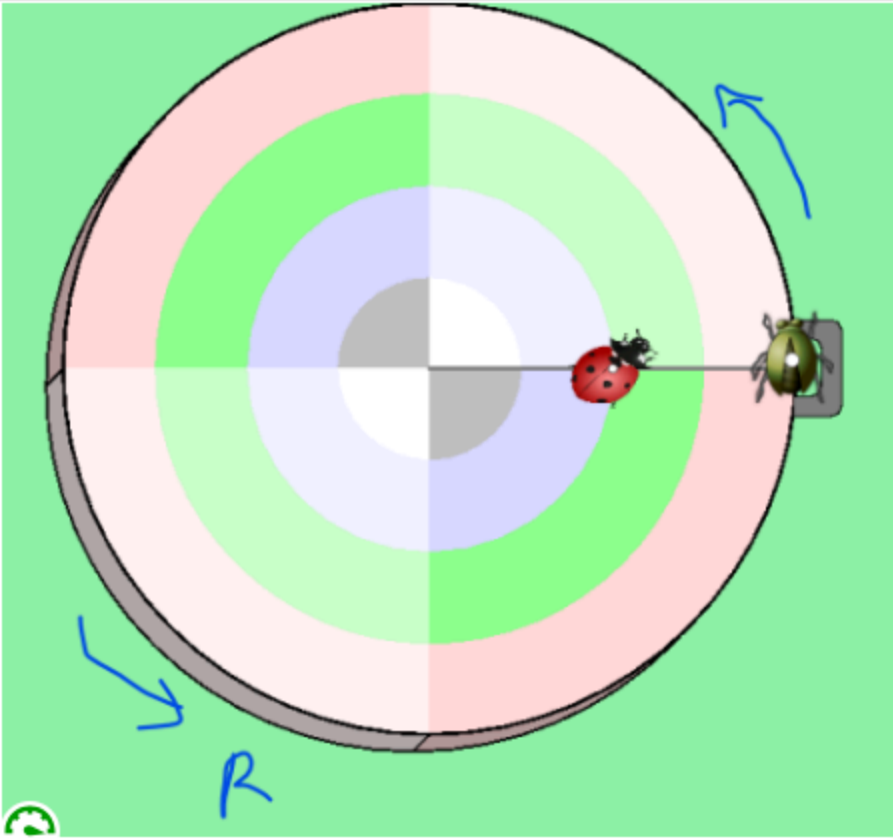
1.4 m/s @ 32°/+x-axis

☐

2 m/s @ 32°/+x-axis

⋮

Question 4 20 pts



A platform is rotating at a rate of 90 degrees / s. Its a uniform circular motion (obviously). What is true

(the (linear) speed is (total distance)/time (m/s)

☐

they have the same speed

☐

The speed of the lady bug is twice the speed of the fly

☐

The speed of the fly is 4 times the speed of the lady bug

☐

The speed of the fly is twice the speed of the lady bug

⋮

Question 5 25 pts

Building on the previous question. if the rotational speed is now 270 degrees/ sec (increases by a factor of 3).

The centripetal acceleration of the bugs increases by a factor of ?

hint: so the speed of the bugs is multiplied by 3. The ac is multiplied by ?

$$ac = (\text{speed})^2/R$$

☐

4

☐

9

☐

3

☐

stays the same

Not saved

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